

GAS SENSOR

How does an MQ-6 gas sensor work and how can I connect it in the gas leakage project I am working on?

Ankit

EFY. The output voltage of MQ-6 gas sensor varies depending on the concentration of the gas in the air. In general,



Fig. 1: MQ-6 gas sensor

the output voltage decreases as the concentration of the target gas increases. The sensor's output voltage can be read using an analogue-to-digital converter (ADC) and then processed by a microcontroller to determine the concentration of the target gas.

The sensor output voltage also varies depending on the temperature and humidity of the environment. So, it is important to calibrate the sensor in the specific environment where it will be used. Also, the sensor should be heated for a few minutes before using it to get an accurate reading. An image of the MQ-6 is shown in Fig. 1.

MQ-6 module has four pins: VCC, GND, AO, and DO. VCC is the positive power supply pin, typically connected to a voltage between 2.5V and 5V. GND is the ground pin, connected to the negative power supply. AO (analogue output) pin gives the sensor's output voltage, which can be read using an ADC. DO (digital output) pin can be used to get a digital output from the sensor.

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APP CREATION

I want to create an app for turning a smartphone into a contactless IR thermometer as my project. Where can I get detailed guidance for app creation, please?

Kamal Avula

EFY. You can make an app using MIT App Inventor or Kodular. The code has .aia file in it. When you import the .aia

file in Kodular or MIT App Inventor, you can see how code blocks are attached and coded. Alternatively, you can refer the following link to learn how to make an app easily using code blocks: <http://ai2.appinventor.mit.edu/reference>

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FACE RECOGNITION

Can we use Raspberry Pi 3 instead of Raspberry Pi 4 to make a face recognition robot?

Ehab Pilloor

EFY. Yes, you can use Raspberry Pi 3 in place of Raspberry Pi 4, but the processing will be much slower. For best performance regarding processing the images, use Raspberry Pi 4 with 2GB RAM, but you can experiment with some variations as well.

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TEMPERATURE CONTROL

This is regarding the 'Temperature Controlled Switch' DIY article published in January issue. How should I calibrate LM35 for 60°C instead of 50°C?

Pranjal

EFY. The IC LM35, heart of the circuit, gives variable voltage at the output based on temperature up to 150°C. For every one degree rise in temperature there is 10mV increase in voltage at the output pin. If at 0°C the output of sensor is 0V, when the temperature rises to 10°C the output will become 100mV; at 60°C it will become 600mV.

Calibration of LM35 sensor should be done by heating it. Op-amp LM358 has been used to compare the output voltage of LM35 with the reference voltage. Since you want to calibrate for 60°C, set for threshold voltage of 60°C to trigger the op-amp. You need to set the reference voltage to 0.6V using pot VR1, as at 60°C the output of LM35 will be 0.6V.

As soon as LM35 sensor is heated (up to 60°C), the temperature at pin

3 of IC3 goes higher than the reference voltage at pin 2, so the output of op-amp IC3 at its pin 1 goes high and relay RL1 energises. Your calibration is done.

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SMART WATCH PROJECT

This is regarding the DIY article 'Smart Watch With Fitness Tracking' published in February issue. Is this project open source based?

Mangiferus

EFY. Yes, this project is open source. However open source used in smart watch project is for personal use only as non-commercial activity.

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SMART GLASSES

This is regarding the 'Smart Glasses Like Google Glass Made With OLED Display' article published in November issue. Could you please guide me further on this project as I am very new to Raspberry Pi and programming?

Abhishek

EFY. You can discuss your doubts on <https://www.electronicsforu.com/forums> and learn how to use Raspberry Pi with small projects available on [electronicsforu.com](https://www.electronicsforu.com).

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PCB DESIGN

My teacher has asked me to draw the printed circuit board diagram for my project. Please guide me how I can do so?

Pramod Kumar

EFY. You can design PCBs using software like EasyEDA, Altium, Eagle, Kicad, etc. For the basics, please check <https://www.electronicsforu.com/technology-trends/learn-electronics/printed-circuit-board-pcb-basics>

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